In [1]:
```
import cv2     # Pragya Singhal, CU24220050 ,43
import matplotlib.pyplot as plt

# Load image in grayscale
image = cv2.imread("img1.jpeg", cv2.IMREAD_GRAYSCALE)

# Initialize SIFT detector
sift = cv2.SIFT_create()

# Detect keypoints and compute descriptors
keypoints, descriptors = sift.detectAndCompute(image, None)

# Draw keypoints on the image       # Pragya Singhal, CU24220050 ,43
sift_image = cv2.drawKeypoints(
    image,
    keypoints,
    None,
    flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS,
)

# Display result        # Pragya Singhal, CU24220050 ,43
plt.figure(figsize=(8, 8))
plt.imshow(cv2.cvtColor(sift_image, cv2.COLOR_BGR2RGB))
plt.title(f"SIFT Keypoints Detected: {len(keypoints)}")
plt.axis("off")
plt.show()
from skimage import exposure
from skimage.feature import hog

# Compute HOG features and visualization image     # Pragya Singhal, CU24220050
hog_features, hog_image = hog(
    image,
    orientations=9,
    pixels_per_cell=(8, 8),
    cells_per_block=(2, 2),
    visualize=True,
    channel_axis=None,
)

# Rescale histogram for better visualization
hog_image_rescaled = exposure.rescale_intensity(hog_image, in_range=(0, 10))

# Display HOG visualization
plt.figure(figsize=(8, 8))
plt.imshow(hog_image_rescaled, cmap="gray")
plt.title("HOG Feature Orientation Patterns")
plt.axis("off")
plt.show()
# Load two similar images (e.g., query and train images)
img1 = cv2.imread("img1.jpeg", cv2.IMREAD_GRAYSCALE)
img2 = cv2.imread("img2.jpeg", cv2.IMREAD_GRAYSCALE)

# Find keypoints and descriptors with SIFT
kp1, des1 = sift.detectAndCompute(img1, None)
kp2, des2 = sift.detectAndCompute(img2, None)

# Use Brute-Force Matcher to find matches
bf = cv2.BFMatcher(cv2.NORM_L2, crossCheck=True)
```

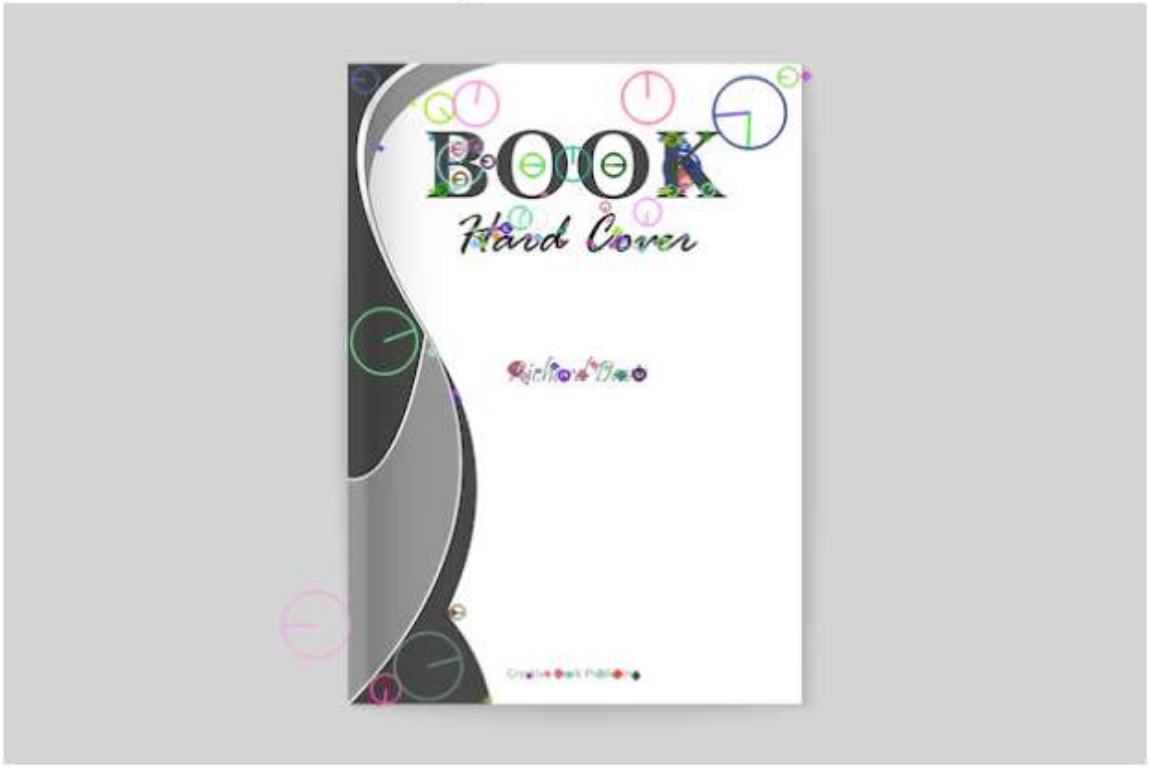
```
matches = bf.match(des1, des2)

# Sort matches based on distance (best matches first)
matches = sorted(matches, key=lambda x: x.distance)

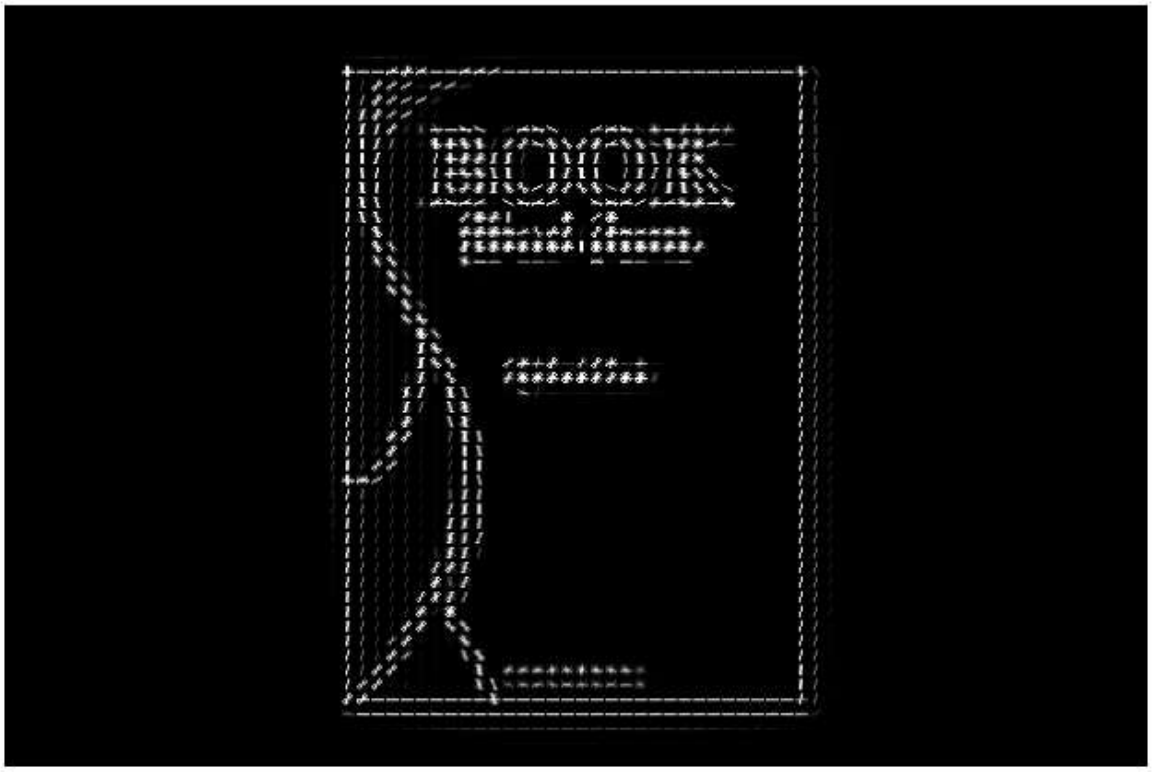
# Draw top matches     # Pragya Singhal, CU24220050 ,43
matched_image = cv2.drawMatches(
    img1, kp1, img2, kp2, matches[:50], None, flags=cv2.DrawMatchesFlags_NOT_DRA
)

plt.figure(figsize=(12, 6))
plt.imshow(cv2.cvtColor(matched_image, cv2.COLOR_BGR2RGB))
plt.title("Top SIFT Feature Matches")
plt.axis("off")
plt.show()
```

SIFT Keypoints Detected: 141



HOG Feature Orientation Patterns



Top SIFT Feature Matches

